

Failure Detection and Recovery of IP Virtual Network Slice using P2MP DSCM Pluggable Transceivers for Xhaul Networks Controlled by Multi-layer SDN Control Framework

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Abstract *This demonstration showcases autonomous failure detection and recovery of Xhaul IP virtual network slices over P2MP DSCM pluggable transceivers, where the optical controller detects the failure, while the multi-layer SDN framework coordinates the slice migration to a backup path. ©2026 The Author(s)*

Overview

Future Xhaul networks are expected to support high capacity, high availability, and dynamic service management with limited manual intervention. In this context, autonomous failure detection and recovery becomes essential for self-healing network operation, where failures or service-degrading conditions are detected and recovered through coordinated control actions [1].

Moreover, 6G Xhaul networks require dynamic provisioning and lifecycle management of end-to-end (E2E) connectivity services to support changing capacity connectivity demands across access, aggregation, metro, and backbone transport segments. In this work, an E2E connectivity service refers to the transport connectivity established across the packet and optical domains, where IP virtual link is supported by the corresponding optical path resources. Such demands may evolve due to application requirements, network conditions, and traffic dynamics.

In Xhaul networks, transport connectivity is required between the central office and multiple edge or cell-site endpoints. This connectivity can be realized through dedicated point-to-point (P2P) optical connections, where each endpoint is served by a separate optical path, or through a shared point-to-multipoint (P2MP) approach enabled by digital sub-carrier multiplexing (DSCM).

Coherent pluggable transceivers are enablers for Xhaul evolution, as they allow IP routers to generate coherent optical signals while supporting traffic aggregation [2]. In this work, an IP router equipped with DSCM pluggable transceivers is referred as IPoWDM router. A P2MP configuration allows a hub pluggable to aggregate traffic from multiple lower-rate leaf pluggables.

This makes the approach suitable for Xhaul scenarios where traffic from different cell-site or edge endpoints is transported over shared optical infrastructure [3]. However, this resource is now involved in optical path planning, pluggable configuration, router configuration, and IP service provisioning. Therefore, coordinated multi-layer control is required to prevent fragmented operation across the IP and optical layers [4].

Furthermore, continuous optical telemetry and event notifications are essential for autonomous recovery. By providing visibility into optical paths, pluggable status, and service conditions, they enable the control framework to detect impairments, trigger recovery actions, and support self-healing IP-optical network operation [5].

Resilient P2MP Xhaul operation also requires protection across the complete P2MP optical path including both the trunk-fibre segment between the central office and the splitter, and the drop-fibre segment between the splitter and the cell-site side as shown in figure 1. Therefore, full protection requires redundant optical resources on both sides of the P2MP system, including backup hub and backup leaf connectivity, so that the affected service can be restored through an alternative optical path.

In our previous demonstration, the multi-layer control framework was used to automate DSCM pluggable discovery, pluggable configuration, and IP virtual slice provisioning [3]. This paper focuses on the next step: autonomous failure detection and recovery of an active Xhaul IP virtual network slice. To address this requirement, we present an autonomous recovery mechanism for an Xhaul IP virtual network slice provisioned over P2MP DSCM pluggable transceivers.

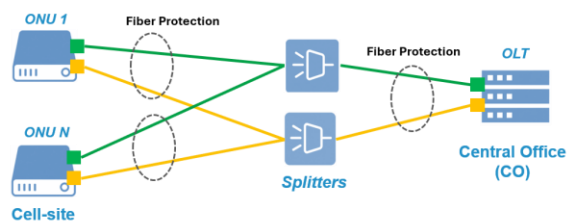


Fig. 1: Full protection scenario in P2MP optical path including trunk and drop fiber protection.

Innovation and Objective

The main innovation of this demonstration is the automated failure detection and recovery of an Xhaul IP virtual network slice over a P2MP optical data plane using DSCM pluggable transceivers. Upon detection of a fiber-cut-like failure affecting the active path, the system detects optical power loss and/or PRE-FEC BER degradation, identifies the affected endpoint, and restores the IP virtual network slice through a backup leaf path.

The control architecture builds on a dual Southbound Interface (SBI) management model, where both the IPoWDM and optical controllers have read access to DSCM pluggable transceiver state through the router, while write access is restricted to the IPoWDM controller [6]. The recovery process is coordinated by the E2E controller. The optical controller supports OTSi service re-planning through an ONF TAPI-based interface [7], while the IPoWDM controller reconfigures the DSCM pluggables through NETCONF (Open-Config) [8], augmented to support digital sub-carrier multiplexing [9], and updates the router

service configuration using the IETF L3VPN service model.

ECOC Relevance

Digital sub-carrier multiplexing (DSCM) is gaining attention in Xhaul networks for its ability to aggregate traffic efficiently over shared optical infrastructure. By splitting a coherent optical signal into independently managed sub-carriers, DSCM enables a point-to-multipoint (P2MP) architecture where a hub pluggable connects to multiple lower-rate leaf nodes. This improves optical resource utilization and lowers deployment costs compared to traditional point-to-point links. However, DSCM also introduces management challenges in sub-carrier allocation, configuration control, resource coordination, and performance monitoring. As a result, open and interoperable management of DSCM-enabled pluggable transceivers is becoming increasingly important for operators and vendors.

This demonstration focuses on the next step after service provisioning: failure detection and recovery of an active IP virtual network slice. In addition to configuration, DSCM-capable pluggable transceivers expose embedded optical monitoring information, such as optical power and performance status, which is being collected and integrated into the control and orchestration framework. The optical controller uses this information to detect optical power loss, identify the affected leaf endpoint, and notify the E2E orchestrator. The E2E orchestrator then coordinates OTSi

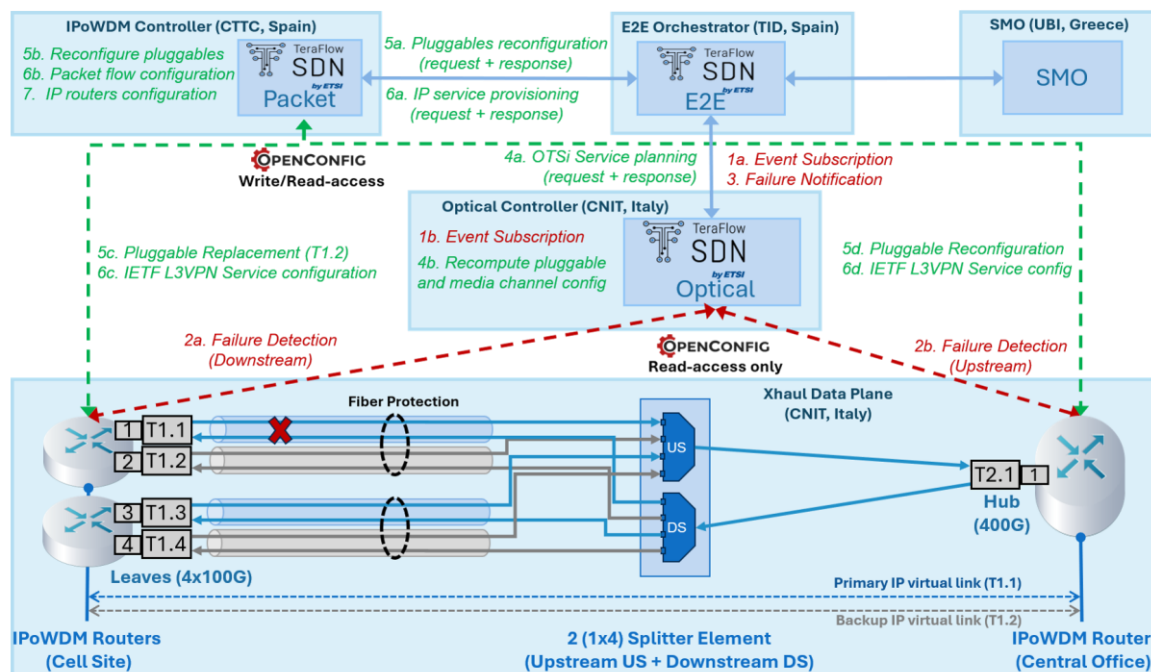


Fig. 2: Distributed testbed view of the demo scenario, including SMO, E2E controller, IPoWDM controller, optical controller and Xhaul data plane elements such as routers, splitter and coherent pluggable transceivers.

service replanning and restores the IP virtual network slice through a backup leaf path activated after failure detection. This demonstrates how DSCM pluggable telemetry, multi-layer SDN control, and orchestration can support autonomous service restoration in resilient IP-optical Xhaul networks. The demonstration is relevant for the ECOC community, especially in the context of resilient IP-optical convergence, DSCM pluggable transceiver management, and autonomous optical networking.

Demo Testbed Setup

A live demonstration will be performed over a remote distributed testbed as illustrated in Fig. 2. The testbed interconnects the physical laboratories of the partners involved. The optical equipment and emulators, packet router, and optical SDN controller are deployed at CNIT premises in Italy [3]. The IPoWDM SDN controller is deployed at CTTC in Spain [10], the E2E SDN orchestrator is deployed at TID in Spain [4], and the SMO layer is deployed at UBI in Greece [11]. All sites are interconnected through a legacy VPN infrastructure to enable coordinated multi-layer control across the distributed testbed.

The optical data plane in the experimental testbed includes commercial DSCM pluggable transceivers in a point-to-multipoint configuration. The IPoWDM router function is implemented using two Edgecore white-box packet devices, which host the DSCM pluggables and provides 100G Ethernet client interfaces toward the cloud client domains. A DSCM pluggable, T2.1, acts as the hub at the central-office side, while four lower-rate pluggables, T1.1 to T1.4, act as leaves at the cell-site side. The hub aggregates traffic flows from the leaf pluggables through two 1×4 splitter elements, supporting upstream and downstream transmission directions. The low-level configuration of the pluggables is performed through direct access to Common Management Interface Specification (CMIS) registers. The configuration parameters are represented through an augmented OpenConfig terminal-device YANG model to support the configuration of digital sub-carriers [8].

Demo Implementation and Workflow

On top of the control stack, the SMO layer initiates and manages the end-to-end service lifecycle. Below the SMO, the E2E SDN orchestrator governs the converged Xhaul control plane and coordinates two domain controllers: the optical controller and the IPoWDM controller. The optical controller is responsible for optical telemetry collection, pluggable discovery, failure detection, and OTSi service planning/replanning through an ONF TAPI-based interface. In this demo, OTSi planning computes DSCM pluggable and media-channel parameters for provisioning and

recovery. The IPoWDM controller configures the DSCM pluggable transceivers and the IPoWDM router, including the IP-layer virtual slice. The controllers in the multi-layer architecture are based on the ETSI TeraFlowSDN (TFS) controller and are enhanced to support optical network monitoring, alarm subscription, IP virtual link management, DSCM pluggable control, and recovery of the affected IP virtual network slice [4].

The demonstration will be presented to the audience through a live session illustrating the failure detection and recovery workflow as indicated by the annotations in Fig. 1. The workflow assumes that the IP virtual network slice has already been provisioned over the primary leaf pluggable, T1.1, following the service provisioning procedure demonstrated in the previous work [3]. The optical controller continuously monitors the active optical path using metrics obtained from the transceiver pluggables such as received optical power and Pre-Forward Error Correction Bit Error Rate (PRE-FEC BER). During the live demonstration, a fiber-cut-like loss of power condition is manually introduced on the active primary path to emulate a failure scenario.

The failure is detected by the monitoring module of the optical controller which includes the following services. The Telemetry service collects the optical monitoring stream of the metrics “received optical power” and “PRE-FEC BER” using YANG data model paths. The Analytics service validates whether the received optical power and PRE-FEC BER measured values remain within the acceptable operating range. If the monitored values exceed the configured thresholds, the Analytics service generates an optical link failure alarm. The Automation service captures this alarm, identifies the affected endpoint, and notifies the E2E controller via the NBI about the detected failure.

After receiving the failure notification, the E2E controller initiates the recovery procedure. It first sends an OTSi service replanning request to the optical controller, which recomputes the DSCM pluggable and media-channel parameters for the recovery path. Once the optical controller confirms the replanned configuration, the E2E controller requests the IPoWDM controller to configure the selected backup leaf pluggable, T1.2, and update the IP router configuration accordingly. The IPoWDM controller then deploys the new IP-layer service over the backup leaf path. The confirmation from the IPoWDM controller to the E2E controller completes the reprovisioning of the affected IP virtual network slice and finalizes the recovery process.

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